



# **Security Attacks, Threats, Risks and Their Impact on Security Goals**

Cryptography & Network Security(CSA5110)

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# Cryptography

## Cryptography's Role:

- Encryption — protects confidentiality (AES for symmetric, RSA/ECC for asymmetric)
- Hash Functions — provide integrity checks (SHA-256 detects tampering)
- Digital Signatures & PKI — ensure authentication and non-repudiation

### Technical Measures:

- Strong, up-to-date encryption and secure key management
- Firewalls, IDS/IPS, antivirus, secure configurations
- Timely patching, backups, redundancy, recovery plans

### User & Organizational Measures:

- Strong passwords, MFA, least privilege access
- Employee awareness training and phishing drills
- Clear policies, incident response, regular audits

Prevention through layered defenses is far less costly than recovery after a breach.

# Security Goals: The CIA Triad & Attack Impact

## Confidentiality

protect data from unauthorized disclosure

### Controls:

- encryption
- access control
- least privilege

### Affected by:

- Phishing
- Malware
- SQL Injection
- MITM

## Integrity

ensure accuracy and immutability of data

### Controls:

- hash functions
- MACs
- digital signatures
- secure logging

### Affected by:

- Phishing
- Malware
- SQL Injection
- MITM

## Availability

ensure systems and data are accessible when needed

### Controls:

- backups
- redundancy
- DDoS defenses
- disaster recovery

### Affected by:

- Ransomware
- DDoS attacks

**Consequences:** financial loss, data breaches, service disruption, identity theft, reputational damage, regulatory penalties

# Classification of Security Attacks

## Passive Attacks

— observe data without altering it

- Eavesdropping and traffic analysis
- Hard to detect
- Mitigations: strong encryption, link-layer protection

## Active Attacks

— alter or disrupt data and services

- Masquerade, replay, modification, denial-of-service
- Easier to detect
- Mitigations: authentication, integrity checks, IDS/IPS

# Common Cyber Attacks Overview

## **Phishing**

— social engineering to harvest credentials or deliver malware

## **Malware & Ransomware**

— malicious software that steals data, disrupts services, or encrypts files for ransom

## **SQL Injection**

— injects malicious queries to read or modify databases

## **XSS (Cross-Site Scripting)**

— injects scripts into web pages to hijack sessions or deface pages

## **MITM (Man-in-the-Middle)**

— intercepts or alters communication between parties

## **DDoS & Brute Force**

— overloads services or systematically guesses credentials

# Information Security & Threat Landscape

Information security protects data and systems from unauthorized access, misuse, alteration, disclosure, or destruction. Key concepts:

## Threats

Potential events that can exploit weaknesses (malware, phishing, insider threats, disasters)

## Vulnerabilities

Weaknesses in hardware, software, or human behavior (weak passwords, unpatched systems)

## Risk

The impact when a threat exploits a vulnerability (data loss, service outage)

**The relationship: Threat + Vulnerability = Risk**

# Conclusion:

## **Threats, Vulnerabilities, and Risks Form a Critical Chain**

Threats are potential dangers that exploit vulnerabilities in systems and processes. When a threat successfully exploits a vulnerability, it creates a risk that can result in significant harm.

## **Security Attacks Target the CIA Triad**

Every major cyber attack—from phishing to DDoS—compromises one or more pillars of the CIA Triad (Confidentiality, Integrity, Availability)

## **Diverse Attack Vectors Require Layered Defense**

Passive attacks like eavesdropping and active attacks like malware each demand different mitigation strategies.

## **Cryptography is a Cornerstone, Not a Complete Solution**

Encryption, hashing, and digital signatures are vital tools for protecting confidentiality, integrity, and authentication. However, cryptography alone cannot prevent all attacks.

## **Prevention and Preparedness Are Far More Cost-Effective Than Recovery**

Investing in strong security measures, updates, employee awareness, and disaster recovery plans is cheaper than dealing with breaches, data loss, and reputational damage.

